Statistical Modeling of Wheat Variety Classification Using Geometric Data

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***Abstract*— The study focuses on the classification of seed varieties based on various attributes using machine learning models. The dataset encompasses multiple features like area, perimeter, and asymmetry coefficient, presenting a rich ground for analysis. Employing classification algorithms such as k-Nearest Neighbors (kNN), Random Forest (RF), Support Vector Machine (SVM), Decision Tree, and Naive Bayes, this research aimed to identify the most effective model for accurate seed variety classification. Through rigorous experimentation and evaluation, Support Vector Machine (SVM) emerged as the most promising model, attaining an accuracy of 95.24%. This success underscores the efficacy of SVM in handling intricate datasets and nonlinear relationships among features. Nevertheless, the study suggests avenues for further exploration, including advanced feature engineering, model refinement, and potentially expanding the dataset to improve classification accuracy. This research contributes valuable insights into seed variety classification, offering a foundation for enhanced agricultural practices. The findings highlight SVM's potential as a robust classifier and emphasize the importance of continued research in optimizing machine learning models for precise seed variety classification, thereby enhancing crop management and agricultural efficiency.**

Keywords*— *Seed Classification, Support Vector Machine, Machine Learning Models, Decision Tree, Random Forest.

I. INTRODUCTION

One of the primary crops of the globe, wheat is essential to both global agriculture and food security. It is essential for agricultural research, crop breeding initiatives, and maintaining food quality to identify and classify many wheat types. Because wheat kernels are manually graded according to their physical characteristics, this classification has historically been labour-intensive.

Prospective ways to improve and automate the classification process are provided by developments in machine learning and statistical modelling. Predictive models for wheat variety classification have been developed successfully by using seed datasets [1] [2] containing geometrical features of wheat kernels, such as area, perimeter, length, breadth, asymmetry coefficient, and length of the kernel groove.

In order to forecast and categorize different wheat types, we want to investigate how statistical modelling and machine learning may be applied to the Seeds Dataset, which is a collection of geometric features of wheat kernels. We aim to create automated models that can distinguish wheat types accurately based on their geometrical features by utilizing these cutting-edge methodologies.

This research project involves applying statistical and machine learning algorithms to the real-world applications of methods, such as data pretreatment, model building, and assessment. The results are meant to aid in the improvement of wheat variety classification procedures, leading to breakthroughs in crop management and agricultural practices.

The technique used, the conclusions drawn from the analysis, and the consequences of the generated models for the classification of wheat varieties based on geometrical properties will all be covered in detail in the parts that follow.

II. DESCRIPTION OF DATASET

One well-known dataset that is frequently utilized in the field of machine learning and pattern recognition is the Seeds dataset [1] [2]. It is made up of geometric characteristics that were taken from three different varieties of wheat kernels: Canadian, Rosa, and Kama. The statistical analysis and predictive modeling are based on these geometrical qualities, which include seven numerical attributes. Each of the 210 instances in the dataset's structure represents a sample of wheat kernels.

The following are the key attributes present in the dataset:

i. Area: Represents the area of the wheat kernel, measured in square millimeters.
ii. Perimeter: Denotes the perimeter of the kernel, measured in millimeters.
iii. Compactness: Signifies the compactness calculated as the ratio of area to the perimeter squared.
iv. Length of Kernel: Indicates the length of the kernel, measured in millimeters.
v. Width of Kernel: Represents the width of the kernel, measured in millimeters.
vi. Asymmetry Coefficient: Characterizes the asymmetry of the kernel.
vii. Length of Kernel Groove: Measures the length of the kernel groove, also in millimeters.

These geometric features help to distinguish between the many types of wheat by encapsulating important information about the physical properties of the kernels. These quantitative measurements play a crucial role in statistical analysis and automated classification processes for crop classification, agricultural research, and quality evaluation.

The dataset's primary objective is to use these geometrical traits to build predictive models that can accurately classify wheat kernels into their respective classes. The goal of the project is to use machine learning and statistical modeling tools to automate the classification process. This will progress the domains of food science and agriculture while also streamlining agricultural procedures and improving crop management strategies.
Furthermore, the dataset is a perfect tool for testing with different classification algorithms, assessing model performance, and researching the connection between geometrical features and wheat variety categorization due to

its manageable size and well-defined attributes. It is an invaluable resource for investigating statistical modeling techniques and machine learning algorithms for predictive analysis in agricultural domains due to its accessibility and applicability.

III. METHTHODOLOGY

R [3] was used a lot for modelling and data analysis during the assignment. The R environment was used for statistical summaries, exploratory analysis, model construction, and assessment. R's flexibility acted as the only foundation for the assignment's extensive analytical requirements, guaranteeing a unified and well-rounded approach to the analysis procedure as a whole.

A. Data Preparation

The initial data preparation phase involved reading the Seeds Dataset from the specified path and examining its structure and contents. This dataset comprises 210 instances with 8 attributes representing geometric properties of wheat kernels, such as area, perimeter, compactness, length, width, asymmetry_coefficient and variety, the summary statistics revealed the dataset's statistical characteristics, displaying the range and distribution of values across attributes. Visualizations in the form of histograms provided insights into the distributions of numeric variables. Subsequent checks for missing values indicated a complete dataset without any null or missing entries. Moreover, identification and removal of duplicate rows, followed by the elimination of any remaining rows with missing values, ensured the dataset's cleanliness and readiness for further analysis and modelling. This comprehensive preparation process established a solid foundation, furnishing a clean and complete dataset for subsequent stages of exploratory analysis and predictive modelling.

The Seeds Dataset [1] exhibits no class imbalance concerns, depicting a balanced representation among the wheat kernel varieties Kama, Rosa, and Canadian. With 70 instances for each class, totally 210 instances across the dataset, an equitable distribution exists, eliminating class imbalance worries commonly encountered in classification tasks.



Figure 1: Bar chart of 'Variety' Class

B. Data Analysis

Analysis performed to gain deeper insights into the Seeds Dataset through visualizations and correlation analysis. Histograms were plotted for each numeric attribute to visualize their distributions and understand the range and frequencies of values. The grid arrangement facilitated the simultaneous examination of attribute distributions. Notably, attributes like 'Area' and 'Length' displayed variations in their distributions, showcasing diverse ranges of values.

The correlation matrix [4] was computed to assess the relationships between numeric attributes. The resulting correlation matrix unveiled patterns of correlation or association between different geometric properties of wheat kernels. The subsequent creation of a correlation plot using the 'corrplot' function depicted these correlations graphically. Attributes like 'Width' and 'Length' displayed a positive correlation, indicating a potential relationship between these features. Furthermore, the 'Compactness' attribute exhibited moderate correlations with 'Area' and 'Perimeter'.

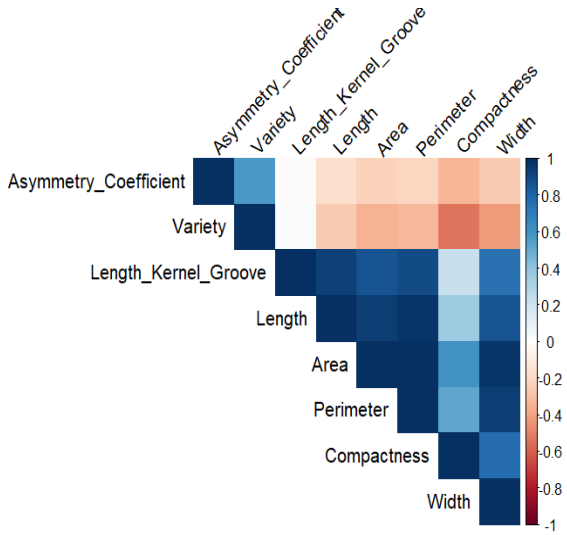


Figure 2: Correlation Matrix

C. Feature Engineering

The feature engineering phase centred on Recursive Feature Elimination (RFE) [5] within the Seeds Dataset. Employing the 'rfe()' function with the Random Forest algorithm ('rfFuncs'), RFE iteratively identified the most influential subset of features for predicting wheat kernel varieties ('Variety' attribute). Utilizing a cross-validation approach with 5 folds, RFE systematically determined the optimal features through the 'rfeControl' function. The resulting output with the optimal feature subset of "Length_Kernel_Groove" "Area" "Perimeter" "Asymmetry_Coefficient", highlighted the selected attributes crucial for classification. Afterwards, a refined dataset ('seeds_data_Re') was formed, comprising solely the identified optimal features alongside the 'Variety' attribute. This curated dataset aims to enhance model efficacy by emphasizing the most discriminative features, streamlining the dataset for improved predictive modelling and classification performance.

D. *Modelling*

The study explores classification models ideal for seed dataset [1] [2] analysis. Robustness to outliers and varied feature distributions prompted the selection of k-Nearest Neighbors (kNN) [6], Random Forest (RF) [7], Support Vector Machine (SVM) [8], Decision Tree (DT) [9], and Naive Bayes (NB) [10] Classification. These models remain adept at interpreting unstructured agricultural data.

i. k-Nearest Neighbors (kNN):

kNN [6] is a straightforward yet effective algorithm used for classification. It assigns classes to new data points based on the classes of its k nearest neighbors in the feature space. This proximity-based approach often performs well on various datasets, especially when there's no clear separation between classes.

ii. Random Forest (RF):

RF [7], an ensemble learning method comprised of multiple decision trees, aggregates predictions from individual trees to arrive at a final prediction. Its capability to handle diverse feature distributions and its resilience to overfitting makes it an ideal choice for this analysis.

iii. Support Vector Machine (SVM):

SVM [8] is known for its ability to handle complex datasets by finding an optimal hyperplane that best separates classes. Its versatility in handling both linear and non-linear classification boundaries make it an attractive option for this study.

iv. Decision Tree:

Decision Trees [9] use a hierarchical structure to simplify feature relationships, making them interpretable and easy to comprehend. Using nodes and connections in a hierarchical diagram, it streamlines the link between the characteristics. Each path starts at the root and is followed by a data-separating sequence until a class at the leaf node is reached. However, they may be prone to overfitting with smaller datasets.

v. Naive Bayes:

Naive Bayes [10] is a probabilistic classifier based on Bayes' theorem. Despite its simplicity, it often performs surprisingly well in various applications, especially when dealing with text classification or datasets where features are conditionally independent.

vi. Testing and Evaluation

This includes partitioning of the dataset into training and testing subsets to facilitate model learning and independent evaluation. Ensuring consistent class distributions, scaling features, and appropriately encoding the target variable laid the groundwork for unbiased model evaluation, enabling the subsequent application of classification algorithms for predictive analysis.

IV. RESULTS

The dataset showed no significant class imbalance and feature engineering has done through Recursive Feature Selection (RFE) to identify the most influential attributes in seed variety prediction. Modelling utilized k-Nearest Neighbours (kNN), Random Forest (RF), Support Vector Machine (SVM), Decision Tree, and Naive Bayes algorithms.

The kNN model showcased an accuracy of 88.1%. It demonstrated balanced sensitivity (ranging from 85.7% to 92.3%) and specificity (ranging from 86.2% to 100%) across different classes, indicating its proficiency in classifying the wheat kernel varieties. The balanced accuracy ranged from 89.3% to 90.7% [11].

In RF model, it showed improved sensitivity ranging from 85.7% to 92.3% and specificity ranging from 89.7% to 100%, with an accuracy of 90.5%. The balanced accuracy was remarkably high, ranging from 90.9% to 94.8% [11].

The SVM model outperformed the other models with an accuracy of 95.2%. It demonstrated high sensitivity and specificity for class predictions, achieving values between 92.9% and 100% for sensitivity and 93.1% and 100% [11] for specificity across different classes.

The Decision Tree model achieved an accuracy of 92.86%. It showcased good sensitivity and specificity across different classes, providing balanced accuracy ranging from 90.58% to 96.43% [11].

The Naive Bayes model achieved an accuracy of 92.86%. It demonstrated high sensitivity and specificity across different classes, with balanced accuracy spanning from 92.71% to 100% [11].

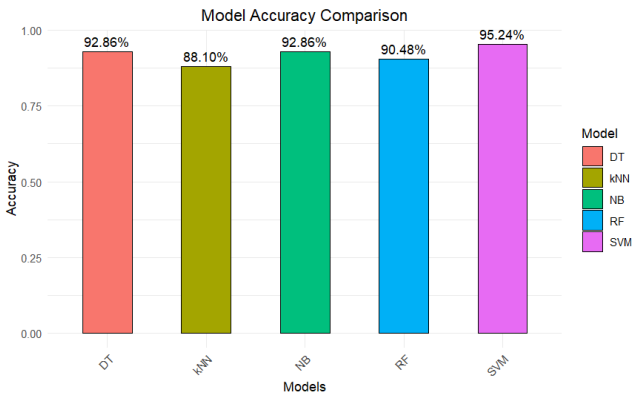


Figure 3: Model accuracy comparison

V. DISCUSSION

The k-Nearest Neighbors (kNN) model has achieved an accuracy of 88.1%. yet, the sensitivity for Class 1 was 92.3%, this indicates that it correctly identified only 92.3% of the Class 1 instances. Meanwhile, the Random Forest (RF) model performed slightly better with an accuracy of 90.48% and higher sensitivity across multiple classes. This

shows RF as a promising model for this dataset due to its balanced performance across classes.

Comparing these models with the Support Vector Machine (SVM) model, which showcased a higher accuracy of 95.24%, but the sensitivity for Class 3 was slightly lower at 92.86%. This demonstrates that while accuracy is higher, SVM might potentially struggle in classifying Class 3 instances compared to the kNN or RF models.

The models SVM have less misclassifications, as can be seen by comparing the confusion matrices; in particular, Class 1, which has the greatest specificity value (93.1%), has fewer misclassifications overall. This implies that, in comparison to other models, SVM may be more successful in accurately identifying cases in Class 1.

The plot (Figure 4) provides a detailed graphical representation of the decision-making process, showcasing node splits, variable importance, and classification rules, aided in understanding the hierarchical structure of the model's predictions.

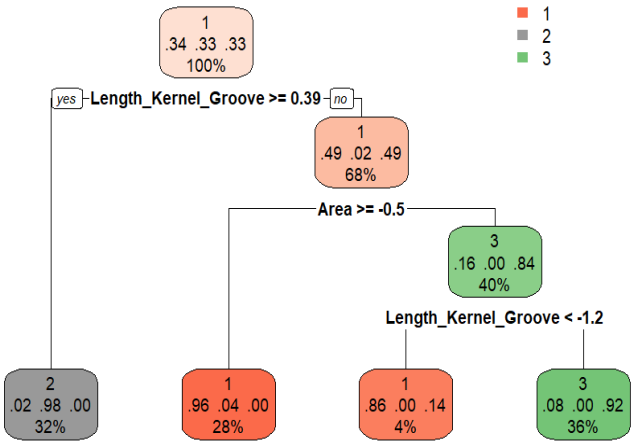


Figure 4: Seed Variety Classification Decision Tree

VI. Conclusion

From the comprehensive analysis conducted on seed variety classification, it is evident that Support Vector Machine (SVM) emerged as the most effective model, achieving a notable accuracy of 95.24%. This success can be attributed to SVM's ability to handle complex datasets and nonlinear relationships between features, making it a robust classifier for this seed variety dataset.

Future work could be performing more advanced feature engineering techniques, or incorporating domain-specific knowledge to enhance model performance. Additionally, investigating ensemble methods or neural networks might yield better classification accuracy. Moreover, expanding the dataset with more diverse samples and attributes could refine the models and potentially address any biases present in the current dataset.

The current analysis produced promising results with SVM outperforming other models, there remains scope for enhancement through advanced feature engineering [12], model refinement, and the inclusion of additional data. This could ultimately lead to more accurate and robust seed variety classification models, contributing to better agricultural practices and crop management.

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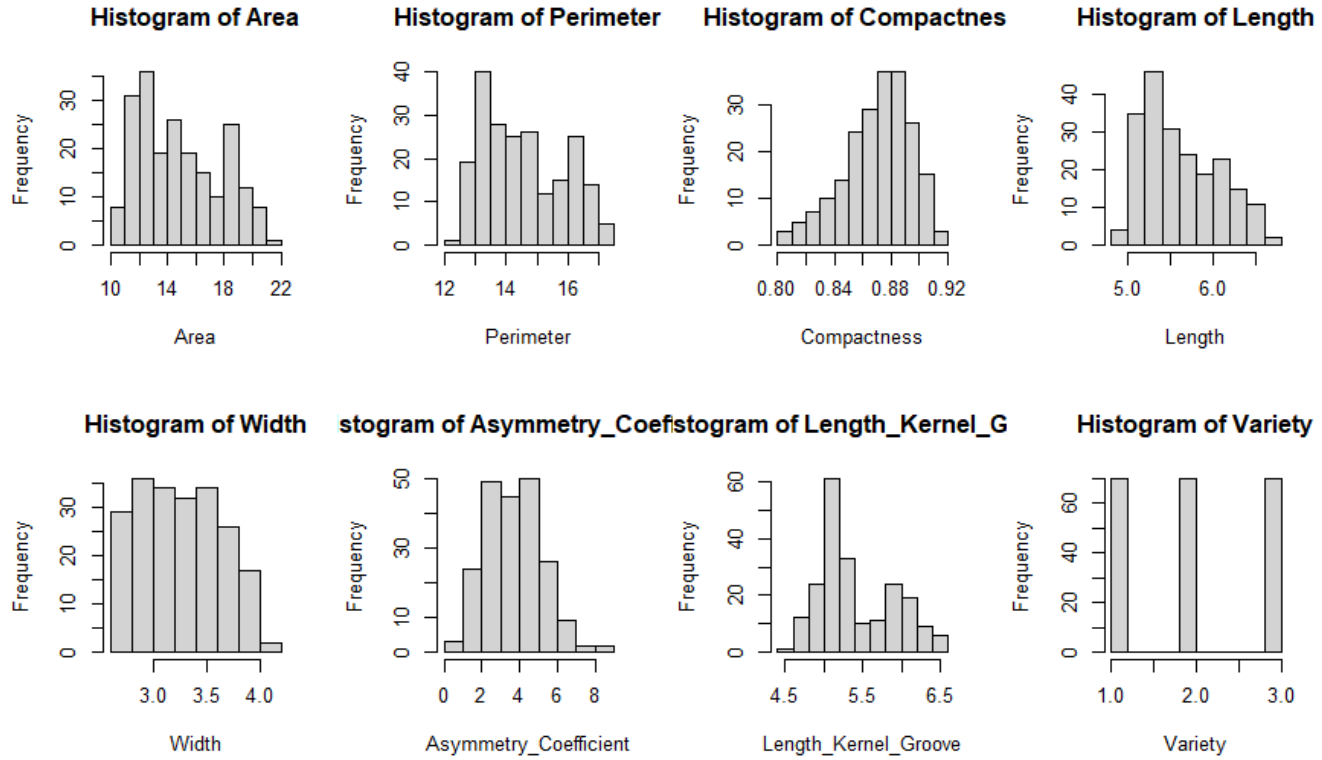
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Appendix A: Distribution Histogram



Appendix B: Outlier Boxplot

